

Geometry of anti-Poisson involutions in the deformations and resolutions of Kleinian singularities

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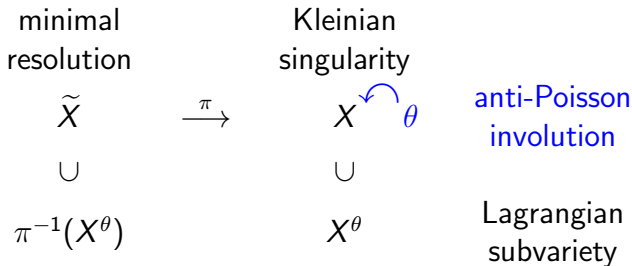
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7 - Lie Theory and Generalizations

- 1 Kleinian singularities
- 2 Anti-Poisson involutions and their fixed point loci
- 3 Preimages of fixed point loci under minimal resolutions
- 4 Deformations and connections to Lie theory

Overview:



Goal: Describe X^θ and $\pi^{-1}(X^\theta)$ as schemes.

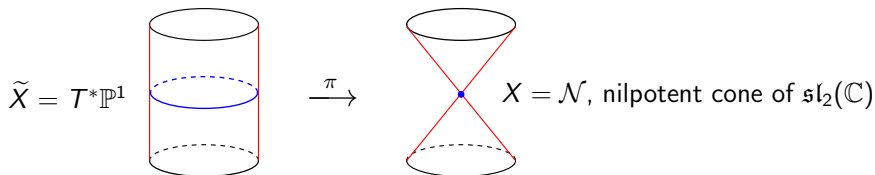
Motivation: Classify (certain) irreducible Harish-Chandra $(\mathfrak{g}, K)$ -modules in geometric terms. Roughly speaking, $\text{Supp}(\text{HC modules}) \approx X^\theta$.

Extended Goal: Replace X by its Poisson deformation X_λ , $\lambda \in \mathfrak{h}/W$, and describe families of Lagrangian subvarieties X_λ^θ and $\pi^{-1}(X_\lambda^\theta)$.

An example

Type A_1 Kleinian singularity: $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^2)$

Anti-Poisson involution $\theta : x \leftrightarrow y$



$X^\theta = \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^2)$ is the union of two $\mathbb{A}^1$'s.

$\pi^{-1}(X^\theta)$ is the union of two $\mathbb{A}^1$'s and a $\mathbb{P}^1$.

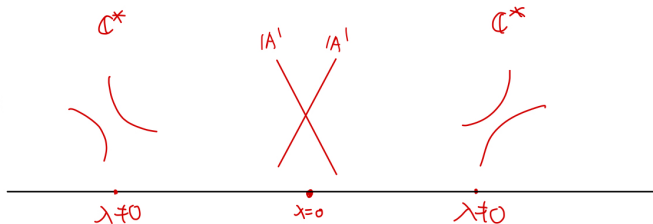
Poisson deformation, $X_\lambda = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^2 - \lambda)$ $\curvearrowright \theta$

$X_\lambda^\theta = \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^2 - \lambda) \simeq \mathbb{C}^*$ if $\lambda \neq 0$.

An example

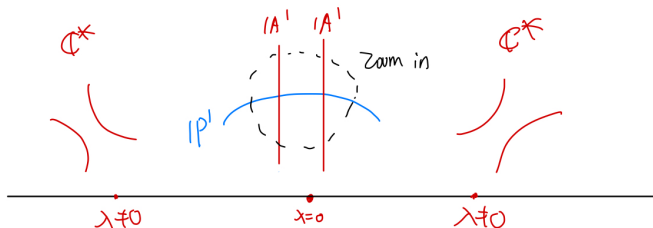
$$X_\lambda^\theta = (X^2 - Z^2 - \lambda)$$

λ

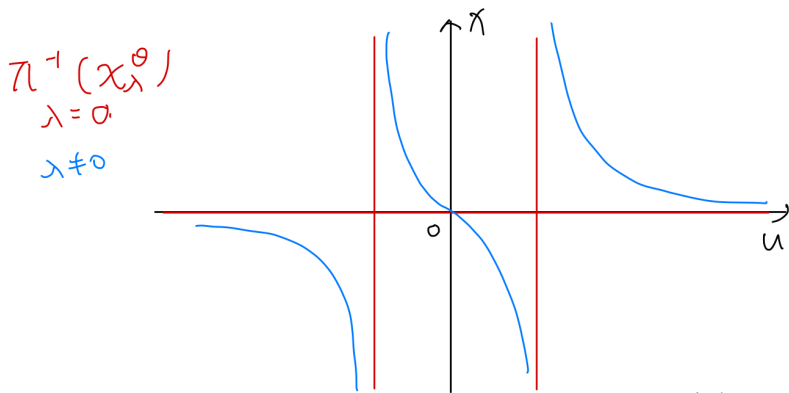


$$\pi^{-1}(X_\lambda^\theta)$$

λ



An example



$$x(u^2 - 1) = 2\lambda u \Rightarrow x = \frac{2\lambda u}{u^2 - 1}$$

$$z = xu - \lambda$$

Kleinian singularities

Let $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$ be a finite subgroup.

The **Kleinian singularities** is the quotient $X := \mathbb{C}^2/\Gamma = \mathrm{Spec} \mathbb{C}[u, v]^\Gamma$.

Example

When $\Gamma = \{\pm I_2\}$, we have $\mathbb{C}[u, v]^\Gamma = \text{even degree polynomials} = \mathbb{C}[x = u^2, y = v^2, z = uv] = \mathbb{C}[x, y, z]/(xy - z^2)$.

Fact (Klein, 1884)

$\mathbb{C}^2/\Gamma \hookrightarrow \mathbb{C}^3$ (one relation), and $\mathbb{C}^2/\Gamma$ has an isolated singularity at 0.

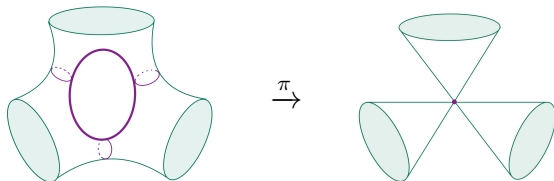
Minimal resolutions

McKay correspondence: Kleinian singularities are in bijection with ADE Dynkin diagrams.

$\pi: \tilde{X} \rightarrow X$, minimal resolution.

$\pi^{-1}(0) = \text{union of } \mathbb{P}^1\text{'s, according dually to ADE Dynkin diagrams.}$

Example:



D_4 singularity
 $x^3 + xy^2 + z^2 = 0$

Anti-Poisson involutions & fixed point loci

Set $X := \mathbb{C}^2/\Gamma$. The algebra of functions $\mathbb{C}[X] = \mathbb{C}[u, v]^\Gamma$ is a graded Poisson algebra with **Poisson bracket**

$$\{f_1, f_2\} = \frac{\partial f_1}{\partial u} \frac{\partial f_2}{\partial v} - \frac{\partial f_1}{\partial v} \frac{\partial f_2}{\partial u}.$$

Definition: An **anti-Poisson involution** of $X = \mathbb{C}^2/\Gamma$ is a graded algebra involution $\theta: \mathbb{C}[X] \rightarrow \mathbb{C}[X]$ such that

$$\theta(\{f_1, f_2\}) = -\{\theta(f_1), \theta(f_2)\}, \quad \forall f_1, f_2 \in \mathbb{C}[X].$$

Definition: The **fixed point locus** is $X^\theta := \operatorname{Spec} \mathbb{C}[X]/I$, where $I = (\theta(f) - f, f \in \mathbb{C}[X])$.

Anti-Poisson involution & fixed-point loci

Proposition 1 (H.)

- There are finitely many anti-Poisson involutions on $\mathbb{C}^2/\Gamma$ up to conjugation by graded Poisson automorphisms.
- The fixed point locus X^θ is reduced.
- If X^θ is not a single point, each irreducible component of X^θ is either $\mathbb{A}^1$ or a cusp.

Example: Type A_n singularity $X = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^{n+1})$. Then θ swapping $x \leftrightarrow y$ is an anti-Poisson involution. We have $X^\theta = \operatorname{Spec} \mathbb{C}[x, y, z]/(xy - z^{n+1}, x - y) \simeq \operatorname{Spec} \mathbb{C}[x, z]/(x^2 - z^{n+1})$, which is a union of two $\mathbb{A}^1$'s when n is odd, a cusp when n is even.

Examples in type A

Example: Anti-Poisson involutions for type A_n Kleinian singularities.

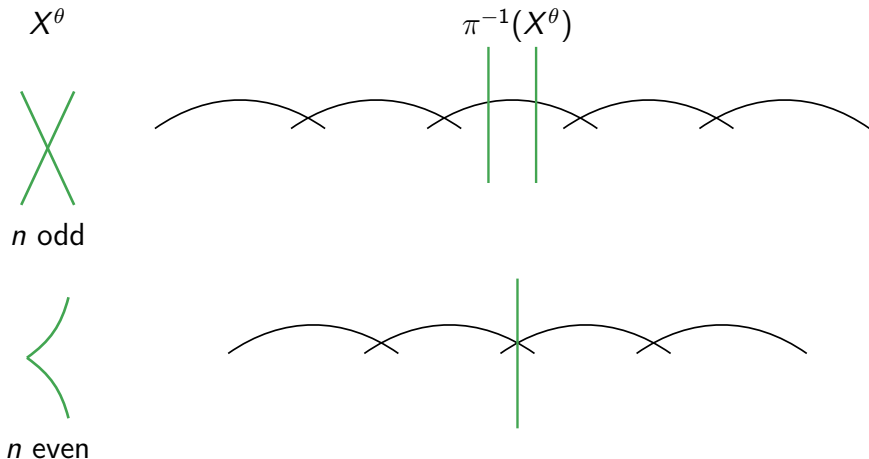
	Type I	Type II	Type III
$A_n, n \text{ odd}$ $xy - z^{n+1} = 0$	$x \mapsto y$ $y \mapsto x$ $z \mapsto z$	$x \mapsto x$ $y \mapsto y$ $z \mapsto -z$	$x \mapsto -x$ $y \mapsto -y$ $z \mapsto -z$
$A_n, n \text{ even}$ $xy - z^{n+1} = 0$	$x \mapsto y$ $y \mapsto x$ $z \mapsto z$	$x \mapsto -x$ $y \mapsto y$ $z \mapsto -z$	
involution of $\mathfrak{sl}_{n+1}$	$M \mapsto -M^t$	$\text{Ad}(I_{\lfloor \frac{n+1}{2} \rfloor, \lfloor \frac{n+1}{2} \rfloor})$	$\text{Ad}(I_{\frac{n+3}{2}, \frac{n-1}{2}})$

Fact: All anti-Poisson involutions come from Lie algebra involutions.

Preimage of fixed point loci under minimal resolutions

Recall $\pi: \tilde{X} \rightarrow X$ minimal resolution. $0 \in X^\theta \Rightarrow \pi^{-1}(0) \subset \pi^{-1}(X^\theta)$.

Example: Consider type A_n singularities with θ swapping $x \leftrightarrow y$.

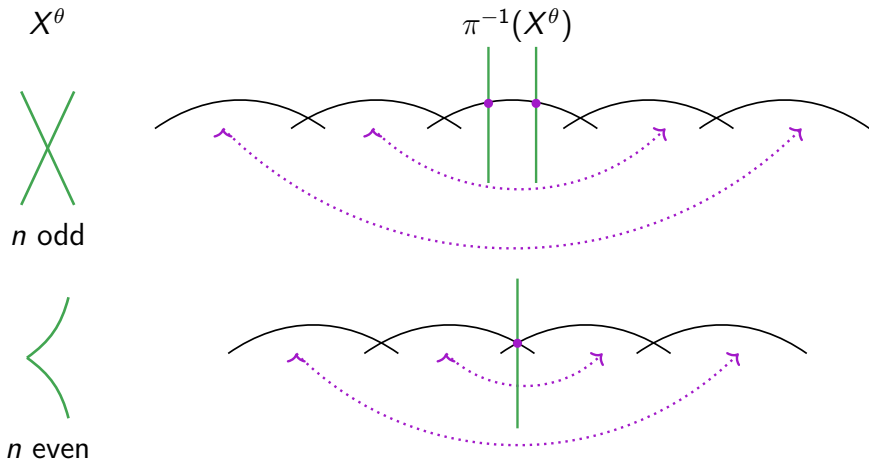


Straight green line is $\mathbb{A}^1$, curly black line is $\mathbb{P}^1$

Preimage of fixed point loci under minimal resolutions

Recall $\pi: \tilde{X} \rightarrow X$ minimal resolution. $0 \in X^\theta \Rightarrow \pi^{-1}(0) \subset \pi^{-1}(X^\theta)$.

Example: Consider type A_n singularities with θ swapping $x \leftrightarrow y$.



Lift the action of θ to the $\mathbb{P}^1$'s

Lift anti-Poisson involutions

$\pi: \tilde{X} \rightarrow X$, minimal resolution, and θ an anti-Poisson involution of X .

Theorem (H., 2025)

There exists a unique anti-symplectic involution $\tilde{\theta}: \tilde{X} \rightarrow \tilde{X}$ such that $\pi \circ \tilde{\theta} = \theta \circ \pi$. It can be constructed via involutions of quiver varieties.

We have $\pi^{-1}(X^\theta) = \pi^{-1}(0) \cup \tilde{X}^{\tilde{\theta}}$.

Fact: $\tilde{X}$ smooth $\Rightarrow \tilde{X}^{\tilde{\theta}}$ is smooth Lagrangian (no intersection, no cusp).

Remark: The preimage $\pi^{-1}(X^\theta)$ is **NOT** reduced in general.

Write $\pi^{-1}(X^\theta) = \sum_{j=1}^m 1 \cdot L_j + \sum_{i=1}^n a_i C_i$ as a divisor, with $L_j \simeq \mathbb{A}^1$, $C_i \simeq \mathbb{P}^1$. There are methods to determine the multiplicities a_i .

Slodowy slice: universal graded Poisson deformation

Let $\mathfrak{g}$ be a simple Lie algebra of types ADE and $\mathcal{N}$ its nilpotent cone. Take $e \in \mathbb{O}_{\text{subreg}}$ and consider the Slodowy slice $S = e + \mathfrak{z}_{\mathfrak{g}}(f)$.

Fact: The Slodowy slice S is the universal graded Poisson deformation of Kleinian singularity $S \cap \mathcal{N} \simeq X$.

Consider the morphism $\varphi: S \rightarrow \mathfrak{h}/W$ defined as the composition

$$\varphi: S \hookrightarrow \mathfrak{g} \rightarrow \mathfrak{g} // G \simeq \mathfrak{h}/W.$$

Zero fiber: $\varphi^{-1}(0) = S \cap \mathcal{N} \simeq X$ is the Kleinian singularity.

Other fibers: $X_{\lambda} := \varphi^{-1}(\lambda)$ is a filtered Poisson deformation of X .

Question: Can we extend the action of θ to X_{λ} , for $\lambda \neq 0$?

Proposition 2 (H.)

θ naturally acts on S and $\mathfrak{h}/W$ by the universal property. The action of θ extends to X_{λ} iff $\lambda \in (\mathfrak{h}/W)^{\theta}$.

Lagrangians in deformations of Kleinian singularities

Question: How to describe X_λ^θ and $\pi^{-1}(X_\lambda^\theta)$, for $\lambda \neq 0$?

- **Step 1:** Determine the singular points of X_λ . Each is a Kleinian singularity of smaller type.
- **Step 2:** Identify the local Type of the anti-Poisson involution near each singular point.
- **Step 3:** Describe the local intersection pattern of projective lines and affine curves, and then connect these local pieces to obtain a global picture.

Examples

Example: An A_2 deformation with singularity type A_1

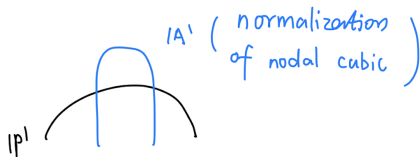
$$\mathcal{X}_\lambda = \text{Spec } \mathbb{C}[x, y, z] / (xy - (z-1)^2(z+2)) \quad \theta : \mathcal{X} \rightarrow \mathcal{Y}$$

$$\mathcal{X}_\lambda^{\text{reg}} = (x^2 - (z-1)^2(z+2))$$

$$z^{-1}(\mathcal{X}_\lambda^{\text{reg}})$$



nodal cubic



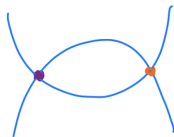
Examples

Example: An A_3 deformation with singularity type $A_1 \times A_1$

$$\chi_\lambda = \text{Spec } \mathbb{C}[x, y, z] / (xy - (z-1)^2(z+1)^2) \quad \theta: X \hookrightarrow Y$$

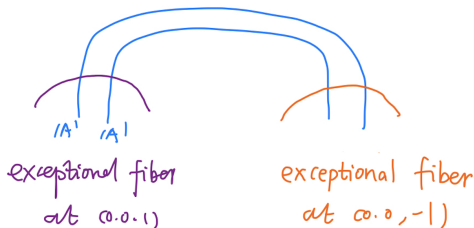
$$\chi_\lambda^\theta = (x^2 - (z-1)^2(z+1)^2)$$

$|A'|$, $x = (z+1)^2$ parabola



$|A'|$, $x = (z-1)^2$, parabola

$$\pi^{-1}(X_\lambda^\theta)$$



Thank you!